ECE 280 Lab 1 – TA Solution Menu
Signals and Systems: Arduino Signal Generator

Use this as a fast grading and troubleshooting guide for:

- Lab 1: Signals and Systems (Arduino Signal Generator)
- Student report template + evidence files

1 Quick Pass/Flag Menu

Pass Immediately If All Are True

- RC filter is physically present and correctly wired: PWM pin → R → node, and C to GND.
- DMM readings are approximately linear at 0%, 50%, and 100% duty cycle.
- Oscilloscope capture shows stable trigger and expected waveform frequency.
- MATLAB plot has 500 samples, labeled axes, and matches scope trend.
- Phase 3 uses timer interrupt ISR (not delay-loop timing), and jitter is lower than baseline.
- Scope proof image includes handwritten student name and ID.

Flag for Review If Any Are True

- Missing physical evidence (scope/DMM/breadboard images).
- Phase 3 code still performs sample timing inside loop with delay calls.
- Scope and MATLAB frequencies disagree by more than 10% without explanation.
- Report includes formulas but no measured values and/or no error analysis.

2 Pre-Lab Answer Key (Expected)

Q1: RC Selection for $f_c \approx 50$ Hz

Formula:

$$f_c = \frac{1}{2\pi RC}$$

Valid examples (commercial values):

- $R = 10\,\mathrm{k\Omega}$, $C = 330\,\mathrm{nF} \Rightarrow f_c \approx 48.2\,\mathrm{Hz}$
- $R = 9.1\,\mathrm{k\Omega}$, $C = 330\,\mathrm{nF} \Rightarrow f_c \approx 53.1\,\mathrm{Hz}$
- $R = 4.7\,\mathrm{k\Omega}$, $C = 680\,\mathrm{nF} \Rightarrow f_c \approx 49.8\,\mathrm{Hz}$

Accept if f_c is roughly 40–60 Hz and derivation is shown.

Q2: RC Impulse/Step Response

Expected forms:

- Impulse response:
$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$
- Step response to 5 V:
$$v_o(t) = 5\left(1 - e^{-t/RC}\right)$$
- 63% time constant:
$$t_{63\%} = \tau = RC$$

Q3: Quantization and SQNR (8-bit)

- Quantization step (0–255):
$$\Delta V \approx \frac{5}{255} = 19.6\,\mathrm{mV}$$
- SQNR for full-scale sine:
$$\mathrm{SQNR_{dB}} \approx 6.02N + 1.76$$

For $N = 8$: $\mathrm{SQNR} \approx 49.9\,\mathrm{dB}$.

3 Lab Measurement Targets

Phase 1 DMM Targets

- 0% duty: ~ 0.00 V (typical ± 0.05 V)

- 50% duty: ~ 2.50 V (typical 2.45–2.55 V)
- 100% duty: ~ 5.00 V (typical 4.95–5.05 V)

Suggested acceptance: within 5–10% unless hardware issues are justified.

Phase 2 Frequency and Shape

- Frequency error target: $\leq 10\%$
- Waveform should be stable and periodic with proper trigger.
- MATLAB trace should visually match scope behavior (shape, period, amplitude trend).

Phase 3 Jitter Expectations

- Baseline (delay-based): wider interval spread.
- Interrupt-based: narrower distribution and lower standard deviation.
- Acceptable improvement: $\geq 30\%$ jitter reduction (std-dev basis).
- Strong implementation: $\geq 50\%$ reduction with clear histogram evidence.

4 Code Key Checks (TA Spot-Check)

Delay-Based Baseline (Phase 2/3.1)

- Uses `micros()` logging or timestamp differences.
- Uses `delay()` or `delayMicroseconds()` in generation loop.

Interrupt Phase (Phase 3)

- Timer register setup is present (Timer1/Timer2 acceptable).
- ISR is concise and handles sample update.
- Shared ISR variables are marked `volatile`.
- Sampling path has no blocking delay calls.

Timer Formula Sanity Check

If using Timer1 CTC mode:

$$f_{int} = \frac{F_{CPU}}{\text{prescaler} \cdot (OCR1A + 1)}$$

If LUT size is N samples/cycle:

$$f_0 = \frac{f_{int}}{N}$$

Common correction with $F_{CPU} = 16\,\text{MHz}$, prescaler $= 8$, $N = 256$, $f_0 = 10\,\text{Hz}$:

- Required interrupt frequency: $f_{int} \approx 2560\,\text{Hz}$
- $OCR1A \approx 780$ (not 6249)

5 Frequent Failure Modes and TA Actions

- **Flat output near mid-level only**: likely wrong probe node or RC short.
  Action: verify probe at RC output node and capacitor to GND only.
- **Heavy ripple/noisy analog output**: likely f_c too high or grounding/wiring noise.
  Action: check R/C values, wiring length, and ground routing.
- **MATLAB timeout/missing data**: wrong COM port or baud mismatch.
  Action: confirm 115200 baud, correct port, terminator, and buffer handling.
- **Interrupt code compiles but jitter not improved**: ISR too long, serial inside ISR, or loop timing still active.
  Action: remove serial from ISR, keep ISR minimal, move logging outside ISR.

6 Grading Menu (Aligned to Rubric)

Use this quick decision map while assigning points:

- Pre-Lab (15): full math, correct formulas, sensible component choices.
- Hardware (12): valid RC build + DMM evidence at 0/50/100%.
- Scope Capture (10): clear image, correct scales, $\leq 10\%$ frequency/amplitude error.
- MATLAB Acquisition (10): 500 samples, labeled plot, scope-consistent behavior.
- Phase 3 Interrupt (12): timer ISR implemented + jitter reduction demonstrated.
- Proof Documentation (21): required proof images complete and identifiable.
- Analysis & Errors (20): numerical errors computed + ranked error sources justified.

Suggested Deduction Guide

- Missing one required proof image: -5 to -10 (severity-based)
- No jitter comparison evidence: -4 to -8
- No error calculations: -5 to -10
- Copy/paste code with no measured-data linkage: -3 to -8

7 Minimum Submission for Passing Work

A minimally acceptable submission should include:

- Pre-lab equations and numerical results.
- One valid RC hardware photo and DMM verification table/photo set.
- One scope capture with visible settings.
- One MATLAB 500-sample plot with labeled axes.
- Baseline vs interrupt jitter evidence and brief improvement statement.
- Short error analysis with at least three plausible sources.